



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
27-06-2021

JEE(Main + Advanced) : NURTURE PHASE-I, I(B), II(A) & NURTURE LIVE PHASE-I, I(B), II(A)

PAPER-2

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **26** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

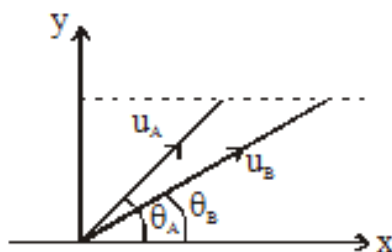
BEWARE OF NEGATIVE MARKING

PHYSICS

SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : -1 In all other cases

1. Two particles A and B are projected from the same point in different directions in such a manner that vertical component of their initial velocity are same then ratio of their horizontal range will be equal to :-



(A) $\frac{R_A}{R_B} = \frac{u_A}{u_B}$

(B) $\frac{R_A}{R_B} = \frac{\tan \theta_B}{\tan \theta_A}$

(C) $\frac{R_A}{R_B} = \frac{\tan \theta_A}{\tan \theta_B}$

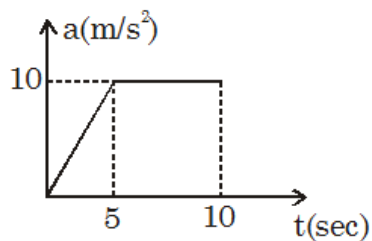
(D) $\frac{R_A}{R_B} = \frac{\sin \theta_A}{\sin \theta_B}$

Space for Rough Work

2. A police jeep is chasing with, velocity of 45 km/h a thief in another jeep moving velocity 153 km/h. Police fires a bullet with muzzle velocity of 180 m/s. The velocity it will strike car of the thief is
(A) 150 m/s (B) 27 m/s (C) 450 m/s (D) 250 m/s
3. In a new system of units called royal system of units, units of length, mass and time are each ten times of respective SI units. The unit of force in this system will be
(A) equal to that in SI system (B) 10 times that of SI system
(C) 100 times that of SI system (D) 1000 times that of SI system
-

Space for Rough Work

4. If at $t = 0$, velocity of particle is 2 m/s , find velocity at $t = 10 \text{ sec}$.



- (A) 100 m/s (B) 73 m/s (C) 77 m/s (D) 70 m/s
5. A lift is moving with constant velocity. A coin is dropped at height 20m from the floor of lift, find the time taken by coin to reach the floor :- (Take $g = 10 \text{ m/s}^2$)
- (A) 1 s (B) 0.5 s (C) 2 s (D) 1.5 s

Space for Rough Work

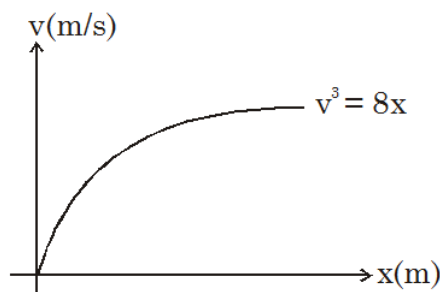
6. The bottom of an electric pole of height 30m is situated at a distance of $30\sqrt{3}$ m from you due 60° North of East. The magnitude of the position vector of a crow sitting on the top of the pole relative to you is
- (A) 30 m (B) $30\sqrt{3}$ m (C) 60 m (D) 50m
7. A ball is projected vertically upward from ground. Select the correct option : (taking upward as positive direction)
- (A) Acceleration is positive while going up and negative while ball is moving down.
- (B) $\vec{v} \cdot \vec{a} > 0$ during upward motion.
- (C) $\vec{v} \cdot \vec{a} > 0$ during downward motion.
- (D) Ball returns to ground with same velocity as velocity of projection.

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

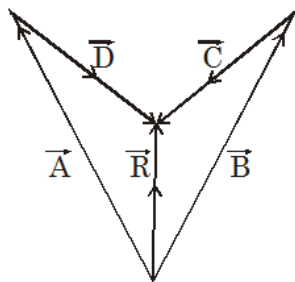
1. The v-x curve for an object moving along the x-axis shown in the diagram :



- (A) The acceleration of particle when $x = 1\text{ m}$ is $\left(\frac{4}{3}\right) \text{ m/s}^2$
- (B) The acceleration of particle when $x = 8\text{ m}$ is $\left(\frac{2}{3}\right) \text{ m/s}^2$
- (C) The acceleration of particle when $v = 2 \text{ m/s}$ is $\left(\frac{8}{3}\right) \text{ m/s}^2$
- (D) The acceleration of particle when $v = 8 \text{ m/s}$ is $\left(\frac{1}{3}\right) \text{ m/s}^2$

Space for Rough Work

2. A dimensionally correct equation is given by $E = \sqrt{\frac{P^4}{2m^2}} + k\sqrt{a}$ in which E is energy, P is momentum, m is mass and k is dimensionless then :-
- (A) Dimensions of a^2 is same as energy
- (B) Dimensions of $\sqrt{a}$ is same as energy
- (C) Dimensions of $(m^2a)^{1/4}$ is same as momentum
- (D) $\frac{E^2}{a}$ is dimensionless
3. Which of the following option(s) is/are correct for the vectors given in figure ?



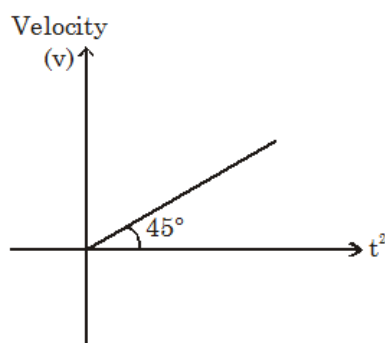
- (A) $\vec{A} = \vec{R} - \vec{D}$
- (B) $\vec{A} - \vec{B} = \vec{C} - \vec{D}$
- (C) $\vec{B} + \vec{C} = \vec{R}$
- (D) None of these

Space for Rough Work

4. A displacement vector of magnitude 10 m has its initial point at (4, 3)m. The y-component of this vector has a magnitude of 6 m. The co-ordinates of the final point of the vector may be
(A) (−4, 9) m (B) (−4, −3) m (C) (12, 9) m (D) (12, −3) m
5. A ball projected vertically upward such that it's speed is same after 2 sec and 6 sec from projection. Then choose the correct options.
- (A) Its time of flight is 8 sec.
(B) Its average velocity is zero from 1 sec to 7 sec.
(C) Its average speed is 20 m/s.
(D) Its average acceleration is zero.

Space for Rough Work

6. Velocity-(time)² graph of a particle moving along x-axis as shown in the figure. Choose the correct options.



- (A) Acceleration of particle at $t = 1$ sec is 2 m/s^2 .
 (B) Particle is slowing down.
 (C) Average speed and magnitude of average velocity of particle are equal.
 (D) Particle is non-uniformly accelerated.
7. A particle is moving along x-axis. Which of the following may be wrong if it is moving towards origin :-
- (A) $\frac{dx}{dt} < 0$ (B) $x \frac{dx}{dt} > 0$ (C) $x \frac{dx}{dt} < 0$ (D) $\frac{dx}{dt} > 0$

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

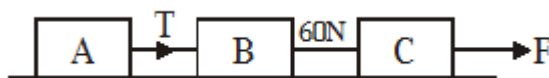
A physical quantity X depends on another physical quantities as $X = Y F e^{-\beta r^2} + Z W \sin(\alpha r)$ where r, F and W represents distance, force and work respectively & Y and Z are unknown physical quantities and α, β are positive constants.

1. If Y represent displacement then $\dim\left(\frac{\alpha Y Z}{\beta F}\right)$ is equal to :-
 (A) $M^{-1}L^2T^2$ (B) $M^{-1}L^2T^{-2}$ (C) $M^1L^1T^{-2}$ (D) None of these
2. If Y represent velocity then $\dim(X)$ is equal to
 (A) ML^2T^{-3} (B) $M^{-1}L^2T^{-3}$
 (C) ML^2T^{-2} (D) None of these

Space for Rough Work

Paragraph for Questions 3 and 4

Three identical blocks each of mass 5kg are connected by ideal strings and are being pulled along a horizontal frictionless table by a constant horizontal force F (refer diagram).



3. What is the magnitude of force F if the tension in the string connecting blocks B and C is 60 N ?
 (A) 60 N (B) 120 N (C) 180 N (D) 90 N
4. Find tension in the string connecting blocks A and B :
 (A) 30 N (B) 20 N (C) 60 N (D) None of these

Space for Rough Work

CHEMISTRY
SEC-1 : (Maximum Marks: 21)

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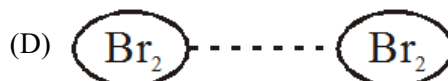
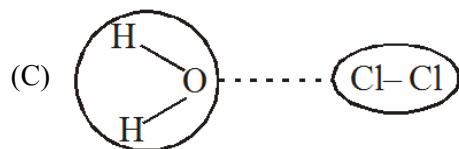
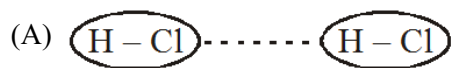
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- Identify the cation having maximum magnetic moment?
 (A) Co^{+2} (B) Cr^{+3} (C) Fe^{+3} (D) Mn^{+4}
- If an electron has spin quantum number $+\frac{1}{2}$ and magnetic quantum number +1 then it cannot be present in.
 (A) d-orbital (B) p-orbital (C) s-orbital (D) f-orbital
- The correct order of increasing electron affinity of the following elements is :
 (A) $\text{O} < \text{S} < \text{F} < \text{Cl}$ (B) $\text{O} < \text{S} < \text{Cl} < \text{F}$
 (C) $\text{S} < \text{O} < \text{F} < \text{Cl}$ (D) $\text{S} < \text{O} < \text{Cl} < \text{F}$

Space for Rough Work

4. In which of the following cases the hydration energy is maximum ?
- (A) Na^+ (B) Mg^{2+}
 (C) Al^{3+} (D) All are having same value
5. Which of the following interaction represents dipole-dipole interaction ?



Space for Rough Work

6. Comment on the C – C bond length for C_2H_6 and C_2F_6 compounds :
- (A) $d_{C-C}(C_2H_6) > d_{C-C}(C_2F_6)$ (B) $d_{C-C}(C_2F_6) > d_{C-C}(C_2H_6)$
(C) $d_{C-C}(C_2F_6) = d_{C-C}(C_2H_6)$ (D) Can't be predicted
7. Identify the incorrect order for size.
- (A) $Sc > Ti > Ti^{+2}$ (B) $Zn > Ni < Cu$
(C) $O^{-2} > Na^{+} > Mg^{+2} < Al^{+3}$ (D) $Ni < Cu < Zn$
-

Space for Rough Work

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- Non-polar molecules are
 (A) CH_4 (B) C_2F_2 (C) C_2F_4 (D) OF_2
- The CORRECT statement is / are :
 (A) NH_3 has lesser dipole moment than that of NF_3
 (B) CO_3^{2-} & SO_3^{2-} is not isostructural
 (C) The shape of XeF_4 is tetrahedral
 (D) PF_5 contains five bonding pairs and zero lone pair of electrons on phosphorous

Space for Rough Work

3. Find the polar and non-planar molecule -
(A) BCl_2F (B) SF_2Cl_2 (C) SiF_3Cl (D) PCl_2F_3
4. In which of the following compounds, lone pair is present in almost non-directional orbital
(A) PH_3 (B) NF_3 (C) AsH_3 (D) H_2O
-

Space for Rough Work

5. Identify the incorrect order -
- (A) $N < O$ (First ionization energy) (B) $Cl > F$ (Electronegativity)
(C) $H_2O < HF$ (Number of lone pair) (D) $O > S$ (Electronaffinity)
6. The orbital is not participated in sp^3d^2 hybridisation
- (A) P_x (B) d_{xy} (C) d_{yz} (D) P_y
7. Which of the following option is/are correct ?
- (A) $N^{3-} > O^{2-} > F^-$ (Ionic radii) (B) $F > N > O$ (Ionization potential)
(C) $Cl > F$ (Electron affinity) (D) $F > Cl$ (electronegativity)
-

Space for Rough Work

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Paragraph for Questions 1 and 2

According to Gillespie and Nyholm, central atom of valence shell e^- pairs arrange them in space in such direction which cause minimum repulsion between them. Because minimum repulsion = minimum potential energy = maximum stability.

Strength of repulsion = $l.p - l.p > l.p. - B.p. > B.p. - B.p.$

1. In sp^3d hybridisation which of the following geometry of the molecule is not possible?

- | | |
|-------------|--------------------------|
| (A) See-saw | (B) T-shape |
| (C) Linear | (D) Distorted octahedron |

2. Which of the following set of molecules/ion have same shape?

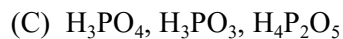
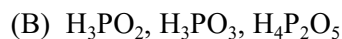
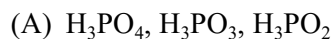
- | | |
|--------------------------|--------------------------|
| (A) SF_4 , BrF_3 | (B) $XeOF_4$, XeF_5^+ |
| (C) ICl_4^- , $TeCl_4$ | (D) XeF_4 , IF_5 |

Space for Rough Work

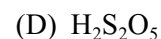
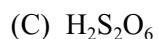
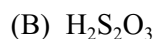
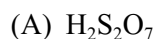
Paragraph for Questions 3 and 4

"Atleast one $P = O$ bond, one $P - O - H$ bond must be exist in oxyacids of phosphorus but $P-H$ bond(s) is/are variable and shows reducing nature with the oxyacids of phosphorus" While such conditions are not applicable with oxyacid of sulphur, boron and nitrogen etc.

3. In which of the following option all the oxyacids of phosphorous can act as reducing agent



4. Which of the following oxyacid of sulphur is dibasic and contain no S-S linkage?



Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 21)

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1. $\log(a+c), \log(a+b), \log(b+c)$ are in A.P. and a, c, b are in H.P., where $a, b, c > 0$. If $a+b = \frac{kc}{4}$, then the value of k is :-
 (A) 2 (B) 4 (C) 6 (D) 8
2. If $a_1, a_2, \dots, a_{15}$ are in arithmetic progression and $a_1 + a_8 + a_{15} = 15$, then $a_2 + a_3 + a_8 + a_{13} + a_{14}$ is equal to :-
 (A) 25 (B) 35 (C) 10 (D) 15
3. Let x_1, x_2, x_3, x_4, x_5 are positive numbers such that $x_1 + x_2 + x_3 + x_4 + x_5 = 15$. If M is maximum value of $(1+x_1)(2+x_2)(3+x_3)(4+x_4)(5+x_5)$ then the value of $\log_6 M$ is
 (A) 0 (B) 3 (C) 5 (D) 1

Space for Rough Work

4. The minimum positive integral value of expression $\frac{x^2 + x + 7}{x + 2}$ ($x \neq -2$)
- (A) 1 (B) 2 (C) 3 (D) 4
5. Sum of all the solution(s) of equation $9^{\log_3(\log_2 x)} = \log_2 x - \log_2^2 x + 1$ is -
- (A) a prime number (B) an irrational number
(C) a rational number which is not integer (D) a composite number
6. If $\log_2 7 = a$ and $\log_3 2 = b$, then the value of $\log_{14} 84$ is -
- (A) $\frac{1 + 3b + 2ab}{ab - b}$ (B) $\frac{1 + 2b + ab}{b + ab}$
(C) $\frac{1 + b + a}{a + b}$ (D) $\frac{1 + b + ab}{a + ab}$
7. The sum of the infinite series $\frac{1}{10} + \frac{2}{10^2} + \frac{3}{10^3} + \frac{4}{10^4} \dots$
- (A) $\frac{10}{81}$ (B) $\frac{1}{9}$ (C) $\frac{1}{8}$ (D) $\frac{17}{22}$

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

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-
1. If $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$ and $M = 6^{15}$, $N = 3^{-100}$ then
- (A) the number of significant digits in M is 11
- (B) the number of significant digits in M is 12
- (C) the number of zeros immediately after decimal and before first nonzero digit in N is 48.
- (D) the number of zeros immediately after decimal and before first nonzero digit in N is 47.
-

Space for Rough Work

2. Which of the following is/are correct ?

(A) $\log_3 5 > \log_{27} 25$

(B) $7^{\sqrt{\log_5 7}} = 5^{\sqrt{\log_7 5}}$

(C) $5^{\sqrt{\log_5 7}} = 7^{\sqrt{\log_7 5}}$

(D) $3^{(\log_9 5)} \cdot 2^{(\log_8 7)} \cdot \sqrt{5} \cdot 7^{2/3}$ is a rational number.

3. For the series $S = 1 + \frac{1}{(1+3)}(1+2)^2 + \frac{1}{(1+3+5)}(1+2+3)^2 + \frac{1}{(1+3+5+7)}(1+2+3+4)^2 + \dots$

(A) 7th term is 16

(B) 17th term is 18

(C) sum of first 10th terms is $\frac{505}{4}$

(D) sum of first 10th term is $\frac{405}{4}$

4. Let T_n and t_n be the n^{th} term of two AP's respectively and ratio of sum of 'n' terms of these AP's is $\frac{2n-3}{4n+1} \forall n \in \mathbb{N}$, then identify the correct option/s

(A) $\frac{T_{10}}{t_{11}} = \frac{7}{17}$

(B) $\frac{T_{10}}{t_{11}} = \frac{17}{7}$

(C) $\frac{t_6}{T_6} = \frac{45}{19}$

(D) $\frac{t_6}{T_6} = \frac{19}{45}$

Space for Rough Work

5. For an A.P. first term 'a' & common difference 'd' both are integers. There is another G.P. whose first term is a & common ratio is d. If 6th term of A.P. is equal to 2nd term of G.P., then possible value of a and d can be -
- (A) a = 6 (B) d = 6 (C) a = 10 (D) d = 2
6. Let $S_n = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots$ upto n terms, which satisfies $S - S_n < 0.001$ where S is sum of given series. If number of terms tends to infinity then number of terms in the series can be
- (A) 8 (B) 9 (C) 10 (D) 11
7. If $a_1, a_2, a_3, \dots, a_{40}$ are in A.P. and $\sum_{r=1}^{20} a_{2r} = 400$ and $\sum_{r=1}^{20} a_{2r-1} = 300$ then choose correct option(s)
- (A) $a_7 - a_6 = \frac{4}{3}$ (B) $a_{10} - a_9 = 5$ (C) $a_1 = -80$ (D) $a_{40} = 115$

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Given that, $A = 2^{\log_3 e^{\log \sqrt{e^3}}}$

$$B = \sqrt{17 + 12\sqrt{2}} - \sqrt{17 - 12\sqrt{2}};$$

and $(\sqrt{2} - 1)^{-C} = \frac{4}{A + B}$

On the basis of above information, answer the following questions :

1. If $4\alpha = A + B$ then the value of $\alpha^3 + \alpha^2 - 7\alpha + 8$ is equal to
 (A) $2\sqrt{2}$ (B) 4 (C) 5 (D) 11
2. For three numbers $\frac{A+B}{4}, \lambda C, \frac{B-A}{4}$ select the **INCORRECT** option
 (A) If they are in G.P. then $\lambda = 1$
 (B) If they are in A.P. then $\lambda = -\sqrt{2}$
 (C) If $\lambda = 1$ then sum of infinite terms finitely exists
 (D) If $\lambda = 1$ then sum of infinite terms is not finite

Space for Rough Work

Paragraph for Questions 3 and 4

A geometric series has first term 'a' and common ratio 'r'. The second term of the series is 4 and the sum upto infinite terms of the series is 25

3. The sum of all possible values of a is -

(A) 5 (B) 15 (C) 20 (D) 25

4. If 'r' is taken the larger of its two possible values then the smallest value of n for which S_n exceeds 24 is -

(A) 12 (B) 13 (C) 14 (D) 15

Space for Rough Work

Space for Rough Work